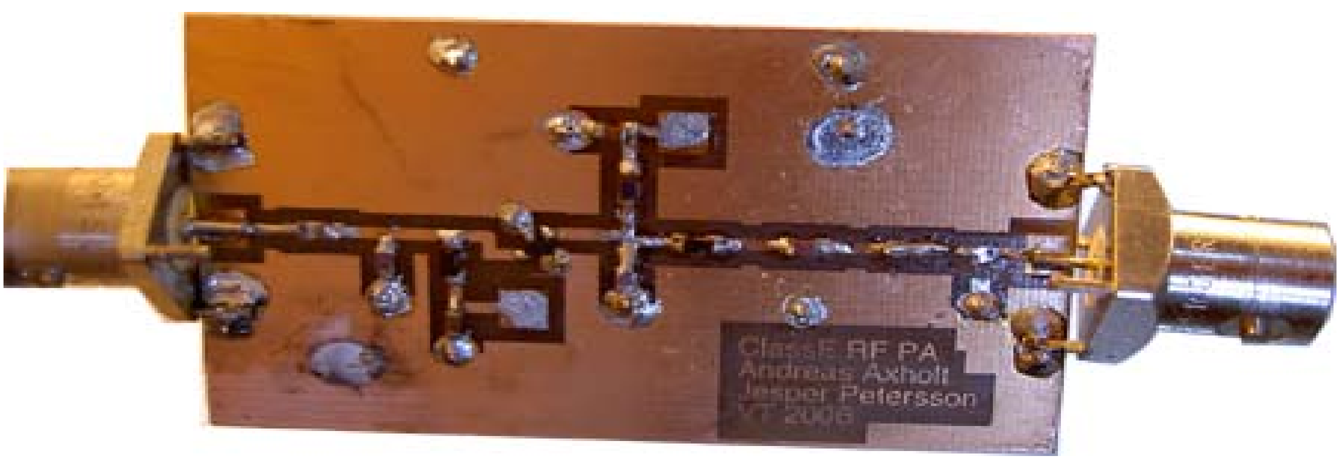
LUNDS TEKNISKA HÖGSKOLA
Lunds universitet

Department of Electroscience
ETI 041 – Radioproject 2006

CLASS E RF PA REPORT



A design of high efficiency Class E power amplifier operating at 100 MHz is presented. The PA utilizes a PHEMT transistor. The design has been investigated both theoretically and practically by simulation (ADS) and implementation. A peak PAE of 74% was achieved at 95.4 MHz with a transducer gain of 19.7 dB.

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2. Introduction

Wireless portable applications have become extremely popular and consumers start to demand longer battery life of their products. Due to the consumers' demand developers has to take power consumption in mind when designing. In a portable transmitting application the majority of the power is consumed in the power amplifier.
The main purpose of this project was to obtain knowledge about Class E amplifiers and where problems arise. During the whole project we did not have a formal specification to fulfill since the topic was new to both of us and our supervisors. The only goal we worked towards was to deliver 20 dBm into a 50 Ohm load and as high efficiency as possible.

3. Overview of Class E amplifier

In June 1975 Sokal published a paper [1] on a Class E amplifier. The main idea behind the Class E topology is to utilize a switching transistor. The reactive circuit at the drain shapes the drain voltage to have both zero value and zero slopes at switch turn-on which ideally will result in 100 % efficiency.

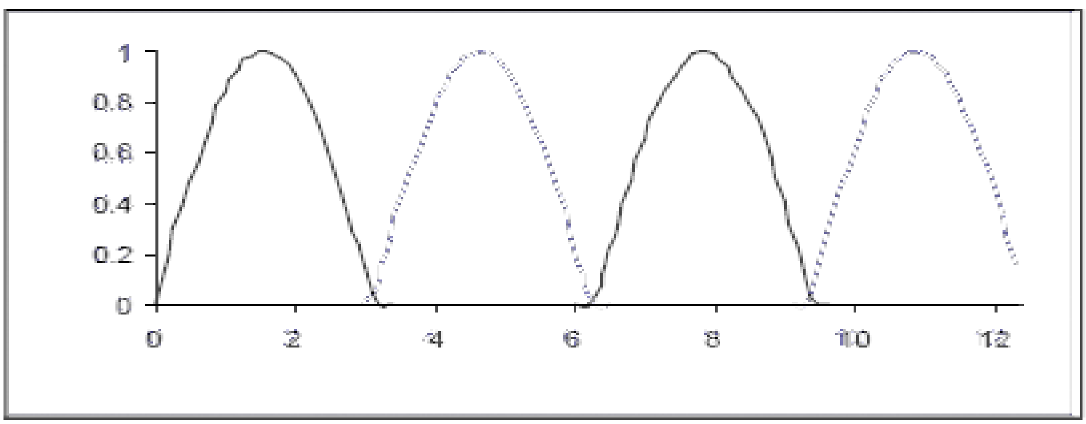

Figure 1. Ideal Drain voltage vs Idrain curves

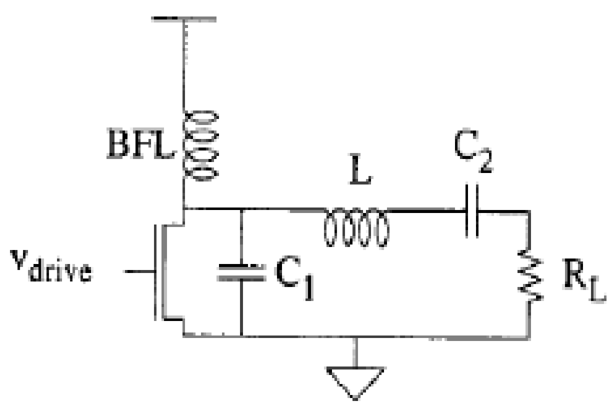

Figure 2. Diagram of Class E amplifier.

The analysis of the Class E amplifiers has been reported in Sokal's paper [1] and is reproduced here.

When the switch is off the voltage v_s is given by solving the equation:

$$C_1 \frac{dv_s}{dt} = I_{ds}(1 - a\sin(\omega_s t + \phi)) \qquad \textit{eq.1}$$

Where ω_s is the signal frequency, I_{ds} is the dc portion of the drain current and the constants *a* and ϕ are yet to be calculated. v_s can be represented as, by integrating eq.1 and setting integration constant equal to cos(ϕ).

$$v(t) = \frac{I_{ds}}{\omega_s C_s}(\omega_s t + a(\cos(\omega_s t + \phi) - \cos(\phi)) \qquad \textit{eq. 2}$$

Optimum operation of a Class E amplifier requires two conditions:

$$\frac{dv_s}{dt}(\frac{T_s}{2}) = 0 \qquad \textit{Cond. 1a}$$

$$v_s(\frac{T_s}{2}) = 0 \qquad \textit{Cond. 1b}$$

Using condition 1a and 1b the constants *a* and ϕ were calculated:
a = 1.86
ϕ = -32.5

Now the voltage v_s and the capacitor's, C_1, current i_s are known in the whole range:

$$v(t) = \frac{I_{ds}}{\omega_s C_s}(\omega_s t + a(\cos(\omega_s t + \phi) - \cos(\phi)) \qquad 0 < \omega_s t < \pi \qquad \textit{eq. 3}$$

$$v_s(t) = 0 \qquad \pi < \omega_s t < 2\pi \qquad \textit{eq. 4}$$

$$i_s(t) = 0 \qquad 0 < \omega_s t < \pi \qquad \textit{eq. 5}$$

$$i_s(t) = I_{ds}(1 - a\sin(\omega_s t + \phi)) \qquad \pi < \omega_s t < 2\pi \qquad \textit{eq. 6}$$

By simply dividing v_s by i_s we obtain the impedance looking out from the transistor's drain in figure 2:

$$Z = \frac{0.28}{\omega_s C_s} e^{j48^0} \qquad \textit{eq. 7}$$

Sokal published a later paper in Janaury 2001 where he reported empirical equations taking the RFC into account, these are also the equations used when designing the output network.

$$C_1 = \frac{1}{2\pi f R(\frac{\pi^2}{4} + 1)\frac{\pi}{2}}(0.99866 + \frac{0.91424}{Q_L} - \frac{1.03175}{Q_L^2}) + \frac{0.6}{(2\pi f)^2 L1} \qquad \textit{eq. 8}$$

$$C_2 = \frac{1}{2\pi f R}\left(\frac{1}{Q_L - 0.104823}\right)\left(1.00121 + \frac{1.01468}{Q_L - 1.7879}\right) + \frac{0.2}{(2\pi f)^2 L1} \qquad \textit{eq. 9}$$

$$L_2 = \frac{Q_L R}{2\pi f} \qquad \textit{eq.10}$$

4. Design procedure

4.1. Finding an appropriate transistor

Class E PAs are widely used by radio amateurs operating at AM frequencies. For this application there is a wide spectrum of amplifiers fulfilling their needs. Switching at 100 MHz puts a rather high demand on switching time, approximately 1 ns which is one fifth of a half 100 MHz period.

4.2. Choosing supply voltage

For the transistor there was a rather limited maximum V_{ds} rating. It was shown in simulation that the V_{ds} during operation varied between 0 volts and V_{dd}*(~4).
To ensure that the transistor acts as switch, i.e completely turns off and on, the gate was biased to its' pinch-off voltage.

4.3. Designing output network

When designing the output network one has four things to keep in mind.

4.3.1.Output power

The output power is defined as $P_{out} = \frac{V_{dd}^{\ 2}}{R_{load}} * 0.566[W]$ *eq.11*

4.3.2. Maximum current through the transistor

When choosing R_{load} one has to make sure that the current does not exceed maximum transistor ratings. Although the datasheet says "Under large signal conditions, V_{GS} may swing positive and the *drain current may exceed* I_{dss}. These conditions are acceptable as long as the maximum P_{diss} and P_{in} max ratings are not." Since there will be variations in components we decided to design for a robust circuit that can handle some discrepancies in the output network.

The current is defined as: $i_d = 1.7 * \frac{V_{dd}}{R_{load}} [A]$ *eq. 12*

4.3.3. Choosing a Q value

The choice of Q_L involves a trade-off among operating bandwidth (wider with lower Q_L), harmonic content of the output power (lower with higher Q_L) and power loss in the parasitic resistances of the load-network inductor L_2 and capacitor C_2 in figure 2.(lower with lower Q_L).

In an article by Sokal, “Harmonic output of Class E RF power amplifiers and load coupling network design”, [2] he reports an imperial equation saying that the second harmonic relative power compared to the main frequency is

$$\frac{P(2\omega_s)}{P(\omega_s)} = (\frac{0.51}{Q})^2 \qquad \textit{eq.13}$$

4.3.4 Load transformation

Our informal specification said that the power should be delivered to a 50 Ohm load. To increase our degrees of freedom when designing for delivered power, maximum current and Q value we inserted an impedance transformation network between the 50 Ohm load and the drain voltage shaping network to realize a virtual R_{load}.

4.4. Designing input matching network

To be able to deliver power from the source to the transistor’s gate we had to create a matching network between 50 Ohm and the transistors input impedance. During this process ADS was unavoidable to use since there was limited data in the datasheet needed to match the input impedance of the transistor. Our goal was to utilize the input matching network as a DC-feed for the gate bias.

5. Design parameter

5.1 Circuit one

Parameter	**Value**	**Unit**
Operating freq	100	MHz
P_{in}	1	dBm
$V_{bias,drain}$	1	Volt
$V_{bias,gate}$	-0,5	Volt
R_{load}	5	Ohm
Q	8	

Circuit is shown in figure 5 in appendix A.

5.2 Circuit two

Parameter	Value	Unit
Operating freq	100	MHz
P_{in}	1	dBm
$V_{bias,drain}$	3.757	Volt
$V_{bias,gate}$	-0,5	Volt
R_{load}	50	Ohm
Q	4.1	

Circuit is shown in figure 6 in appendix A.

6. Results

6.1 Theoretically results

6.1.1 Circuit one

Parameter	Value	Unit
Operating freq	100	MHz
P_{out}(f)	20.46	dBm
P_{out}(2f)	-3,4	dBm
Gain	19.46	dB
PAE	100%	

6.1.2 Circuit two

Parameter	Value	Unit
Operating freq	100	MHz
P_{out}(f)	21.95	dBm
P_{out}(2f)	3.85	dBm
Gain	20.95	dB
PAE	100%	

6.2 Simulated results

6.2.1 Circuit one

Parameter	Value	Unit
Operating freq	100	MHz
P_{out}(f)	17	dBm
P_{out}(2f)	-3,4	dBm
Gain	16	dB
PAE	66%	

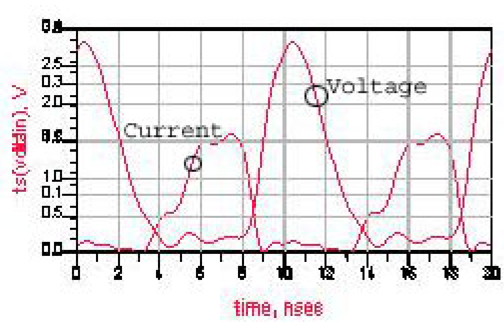

Figure 3. Drain voltage and Drain current

6.2.2 Circuit two

Circuit two was never simulated. Design was based on experiences from circuit one.

6.3 Practical results

6.3.1 Circuit one

Parameter	Value	Unit
Operating freq	95.4	MHz
Pout(f)	20	dBm
P(2f)	-15	dBm
Gain	16	dB
PAE	40%	
BW_{-3dB}	3	MHz

Condition: V_g:-0.5 V, V_{dd}:3.5 V, P_{in}:4 dBm

6.3.2 Circuit two

Parameter	Value	Unit
Operating freq	95.4	MHz
Pout(f)	20.7	dBm
P(2f)	-3.4	dBm
Gain	19.7	dB
PAE	74%	
BW_{-3dB}	3	MHz

Condition: V_g: -0.5 V, V_{dd}:3.757 V, P_{in}:1 dBm

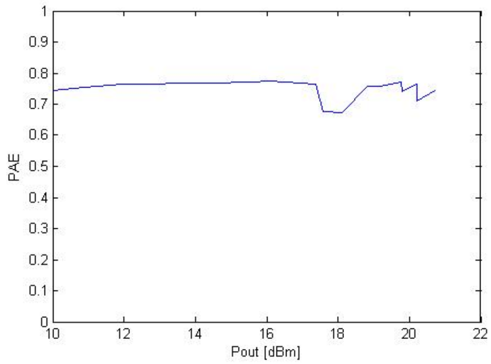

Figure 4. PAE vs Output power

7. Conclusion

Circuit one's design was optimized for an internal load of 5 Ohm. A network was designed to match 5 Ohm with 50 Ohm. During simulation we did not take losses into account which theoretically turned out to be at least 1-2 Ohms, 20-40% of the load it self, in series with the load which clearly reduced the delivered power. Further on we suspected that the output matching network did not operate as it was designed for why a second circuit was designed.
The second circuit was designed for a 50 Ohms load and therefor no output matching network was needed. The removal of output matching network boosted PAE.
It is clearly that circuit number two is more robust since small variation in load impedance will be relatively smaller than for the first circuit where the same variation should be compared with 5 Ohm.

The circuits tended to become unstable for low frequencies, originating from the power supply. This was solved by multiple decoupling capacitors with different frequency characteristics on the supply and bias pads.

Simulations are very useful when the goal is to get an understanding about the circuit. One can easily simulate currents and voltages inside the circuit which are not possible to measure practically. Although, when manufacturing the circuit, many things does not behave as in simulation. This was a part of this project, to get an understanding what happens at higher frequencies, finding the faulting links and how to correct them or work by them.

8. Discussion and personal reflections

This project has been very challenging since neither one of us has come in contact with a Class E amplifier before. There is a clear lack of resources on this topic in literature although the technique is not very new. The main reason for this is because Class E implementations are not suitable for CMOS, which is the main technology edge driving industry, due to its' poor power handling (~0.098 compared to class AB: ~0.32). Additionally the relatively large switch stress does not scale with the trend towards lower power, i.e. lower breakdown voltage, technologies.

The reader might notice the absence of data about intercept point. The data is, according to us, not relevant since it is a switching amplifier topology. The ideal input signal is a square wave with duty cycle 50% which amplitude is equal to $V_{d,sat}$ so that the transistor is fully switched at high input. An additional increase in amplitude should theoretically not influence the delivered power.

An extension to this project could be to add a harmonic trap at the output to minimize harmonics.

9. Acknowledgement

This project is supported by the Department of Electroscience at LTH, Lund University. The authors extend their gratitude to Göran Jönsson, lecturer at LTH, Lund University, Lars Hedenstjerna at LTH, who created our PCBs, Ola Samuelsson and Pär Håkansson, development engineers at Sony Ericsson for their assistance during the project.

References

[1] "Class E RF Power Amplifiers" By Nathan O. Sokal.
[2] "Harmonic Output of Class E RF Power Amplifiers and Load Coupling Network Design" By Nathan O. Sokal and Frederick H. Raab.
[3] "Design of low cost broadband Class E power amplifier using low voltage supply" By Y. Qin, S.Gao, A.Sambell.
[4] "The Design of CMOS Radio-Frequency Integrated Circuits", second edition, By Thomas H. Lee.

Appendix A

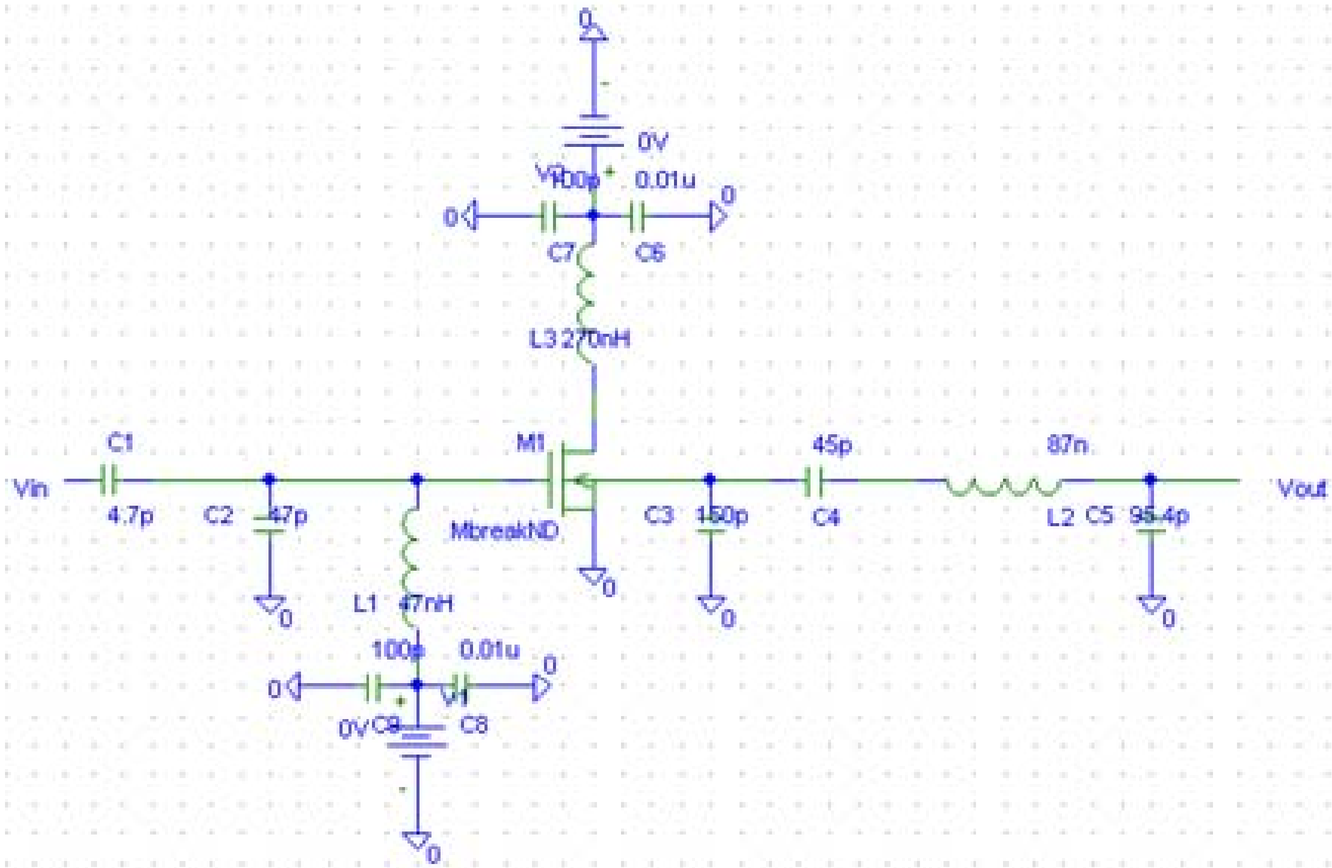

Figure 5. Circuit one

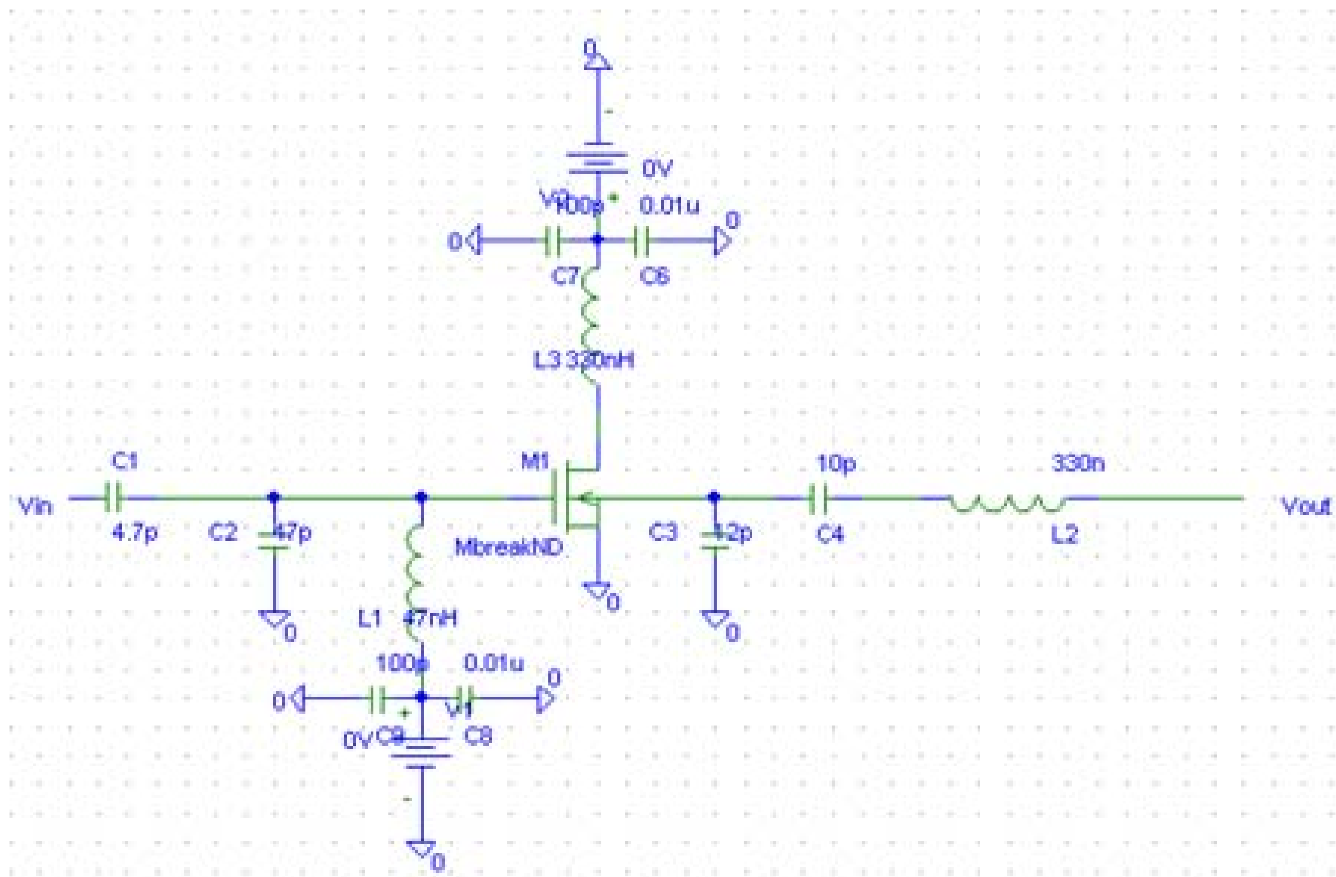

Figure 6. Circuit two